

SOC Threat Detection with Splunk

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1 Introduction

This lab demonstrates how to use a SIEM to detect common cyber attacks. I set up Splunk Enterprise on a Kali Linux VM to act as the central server. I then installed a Splunk Universal Forwarder on my Windows laptop to send Security Event Logs to the SIEM for analysis.

2 Lab Setup

- **SIEM:** Splunk Enterprise (Kali Linux)
- **Endpoint:** Windows 11 Laptop
- **Attacker Tool:** Atomic Red Team (to simulate real attacks)
- **Logs:** Windows Event IDs 4625 (Logon), 4688 (Processes), and 4798 (Groups)

3 Attack Scenarios and Detection

3.1 Scenario 1: Brute Force Attack (T1110.001)

I used a script to try over 100 common passwords against my Windows account. This generated a massive spike in failed login attempts.

Splunk Search Query:

```
index=main EventCode=4625
| stats count by Account_Name, host
| sort -count
```

Result: Detected 9 failed logins in under a minute for the user "Hanna Hahn."

3.2 Scenario 2: Account Discovery (T1087)

I simulated an attacker trying to see which accounts exist on the machine. This is a common "reconnaissance" step.

Splunk Search Query:

```
index=main EventCode=4798 OR EventCode=4799
| stats count by Account_Name, host
```

Result: Splunk showed an automated tool was used to enumerate 6 different local accounts, including the Guest and Administrator accounts.

3.3 Scenario 3: Suspicious Processes (T1059.001)

I ran a simulation of Mimikatz (a tool used to steal passwords). Even though Windows blocked the actual password theft, the "process creation" was still logged.

Result: Detected 21 suspicious child processes coming from `splunkd.exe`. I found that Mimikatz was blocked with an "Access Denied" error (0x00000005).

4 Dashboard and Visuals

I built a simple dashboard in Splunk to track these attacks in real time. It uses bar charts to show which accounts are being targeted and a timeline to see exactly when the Atomic Red Team simulations started.

5 Conclusion and Personal Learning

Through this lab, I learned how to:

1. Configure `inputs.conf` and `outputs.conf` to ship logs between systems.
2. Write SPL to filter through thousands of logs to find specific event codes.
3. Learn the visual work to show attacks in the dashboard.